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Cloud Security with AWS IAM

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Introducing Today's Project!

Project overview

In this project, I will demonstrate how to use AWS IAM to control access and permission setting in our AWS account. I'm doing this project to learn to learn about cloud security from the absolute foundation - every company thinks about access permission and there is a job category called 'IAM Engineers' who focuse on the skills what I am building today.

Tags

What I did in this step

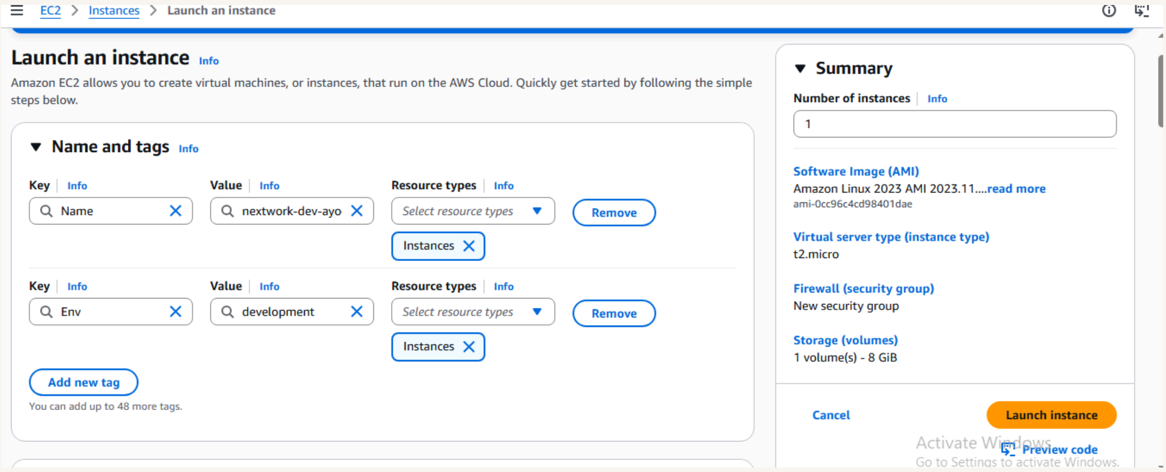
In this step, I will launch two EC2 instances because I need to boost nextwork's computing power as I will be expecting more users and traffic into my website over the summer break.

Understanding tags

Tags are Tags are keywords, labels, or metadata attached to digital content (articles, files, images) to categorize and organize them for easier searching. Unlike rigid folders, tags offer a flexible, multi-dimensional way to classify information by topic or attribute. They are useful for sorting, filtering, and discovering related content across websites, databases, and social media. Tags are used for improved Search and Navigation. Tags are also used for content Organization.

My tag configuration

The tag I've used on my EC2 instances is name and Env.The value I've assigned for my instances are development and production


EC2 > Instances > Launch an instance
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Key Info
Value Info
Resource types Info
Name
nextwork-dev-ayo
Select resource types
Remove
Instances
Key Info
Value Info
Resource types Info
Env
development
Select resource types
Remove
Instances
Add new tag
You can add up to 48 more tags.
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-0cc96c4cd98401dae
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
Activate Windows
Go to Settings to activate Windows.

IAM Policies

What I did in this step

In this step, I will create IAM policy to control the access level to the new user because they should have access to the development environment therefore the development instance and not production environment.

Understanding IAM policies

IAM (Identity and Access Management) Policies are JSON documents that define permissions, allowing or denying access to specific AWS services, actions, and resources for users, groups, or roles. They enforce the "least privilege" principle, controlling what actions (e.g., s3:GetObject) are permitted on resources (e.g., specific S3 buckets) under what conditions

The policy I set up

For this project, I’ve set up a policy using JSON which allows for direct manual editing of policy documents.

Account Alias

What I did in this step

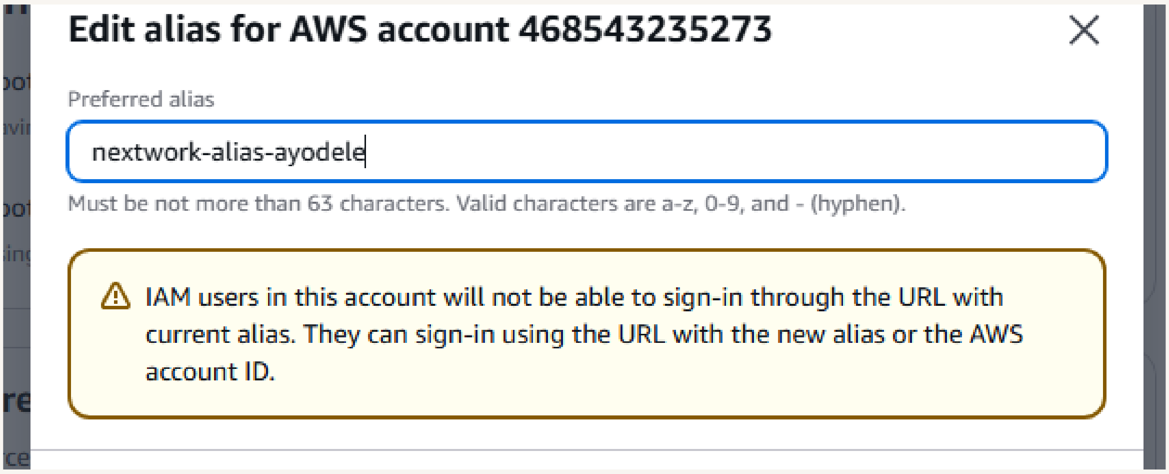
In this step, I will create an AWS account Alias because I want to give my intern access to development instances so that he can do whatever he is permitted to do.

Understanding account aliases

An account alias is is a unique, user-defined name that replaces the 12-digit AWS account ID in your IAM user sign-in URL

Setting up my account alias

Creating an account alias took me about five minutes. Now, my new AWS console sign-in URL is


Edit alias for AWS account 468543235273
Preferred alias
nextwork-alias-ayodele
Must be not more than 63 characters. Valid characters are a-z, 0-9, and - (hyphen).
IAM users in this account will not be able to sign-in through the URL with current alias. They can sign-in using the URL with the new alias or the AWS account ID.

IAM Users and User Groups

What I did in this step

In this step, I will create IAM group for NextWork interns so that I can manage all interns' permission from one place because without this it wiil be difficult to do for individual user.

Understanding user groups

IAM user groups are collections of IAM users designed to simplify permission management by enabling, not disabling, bulk policy assignment. Instead of attaching permissions to individual users, you assign policies to a group, and all members inherit them automatically. They are crucial for organizing, creating consistent security policies, and easing administration for teams

Attaching policies to user groups

I attached the policy I created to IAM group which means that every member of that group automatically inherits the permissions, restrictions, or rules defined in the policy. Instead of assigning policies individually to each user, you apply them at the group level, which has several important effects Consistency: All users in the group follow the same rules, ensuring uniform access control and compliance. Efficiency: Saves time and reduces administrative overhead since you don’t have to manage policies for each user separately. Scalability: As new users are added to the group, they automatically receive the policy without extra configuration. Security: Helps enforce least privilege by ensuring that only the intended group of users has access to specific resources or actions. Simplified Management: Easier to audit and update policies because changes at the group level cascade to all members.

Logging in as an IAM User

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed that some of your dashboard panels are showing access denied alreadyThis was because the user does not have full access to those items as stated in policy.

Testing IAM Policies

What I did in this step

In this step, I will log into Aws using the intern's IAM user because i want to test the intern's access to my production and development instances.

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